

Zaibei Li

Human-Centered Computing
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RESEARCH PROFILE

Doctoral researcher in computer science working at the intersection of multimodal sensing, wearable systems, and human-centered computing. Experienced in designing and building end-to-end systems for multimodal data collection, sensor fusion, interpretable behavioral modeling, and interactive AI applications. Interested in research internship opportunities related to multimodal sensing, wearable systems, and human-centered AI.

EDUCATION

Ph.D. Computer Science, University of Copenhagen, Copenhagen, 2027 (expected)
M.S. Computer Science, University of Copenhagen, 2022
B.S. Information Technology, Hong Kong Polytechnic University, 2019
Exchange Computer Science, The Korea Advanced Institute of Science and Technology, 2017

TECHNICAL SKILLS

Python, C#, Java, PyTorch, TensorFlow, OpenCV, Unity, Docker, Git, IoT prototyping, stream processing

RESEARCH EXPERIENCE

Hiroshima City University

Visiting Researcher

Hiroshima, Japan
Nov 2025–Jan 2026

- Prototyped an egocentric multimodal sensing setup using mobile devices, embedded sensors, and smart glasses for real-world collaborative learning analysis.
- Developed CoLA, a wearable platform with an interactive AI facilitator for real-time collaborative learning analytics; established a research collaboration with Meta Project Aria.

University of Copenhagen

Doctoral Researcher

Copenhagen, Denmark
Feb 2024–Present

- Designed and developed OpenMMLA, a reusable IoT-based toolkit for multimodal data collection, synchronization, and analytics in collaborative learning environments.
- Built interactive pipeline components for sensor fusion, behavioral modeling, and visualization, enabling interpretable analysis of complex group interactions in collaboration with Life Campus on a Novo Nordisk Foundation-funded project.

University of Copenhagen

Research Assistant

Copenhagen, Denmark
Jun 2023–Feb 2024

- Prototyped sociometric badges and AprilTag-based spatial tracking for multimodal learning analytics.
- Built early data collection and processing workflows for mBox, a multimodal sensing prototype that later evolved into OpenMMLA.

SELECTED PROJECTS

CoLA: A wearable multimodal AI platform for egocentric sensing, real-time behavioral analysis, and interactive facilitation in collaborative learning settings, [link](#)

OpenMMLA: An open-source IoT toolkit for multimodal data collection, synchronization, and analytics across real-world collaborative environments, [link](#)

mBox: An early multimodal sensing prototype featuring sociometric badges, audio pipelines, and AprilTag-based spatial tracking; later evolved into OpenMMLA, [link](#)

SELECTED PUBLICATIONS AND PREPRINTS

- 2026 **Z. Li** et al. “CoLA: An Evidence-Continuity System for Multimodal Observation and Support in Collaborative Learning.” *To be resubmitted to Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT), November 2026 round [In revision]*
- 2026 **Z. Li**, V. Holm-Janäs, S. Yamaguchi, and D. Spikol. “Designing for Transparency: Gaze-Augmented Collaborative Action Recognition with Vision-Language Models.” *Proceedings of the 26th IEEE International Conference on Advanced Learning Technologies (ICALT ’26) [Accepted]*
- 2026 **Z. Li**, Q. Li, S. Yamaguchi, and D. Spikol. “BadgeX: IoT-Enhanced Wearable Analytics Meets LLMs for Collaborative Learning.” *arXiv preprint*, arXiv:2604.04093 [Preprint]
- 2025 **Z. Li**, S. Yamaguchi, and D. Spikol. “OpenMMLA: an IoT-based Multimodal Data Collection Toolkit for Learning Analytics.” *Proceedings of the 15th Learning Analytics and Knowledge Conference (LAK ’25)*, <https://doi.org/10.1145/3706468.3706525> [Oral presentation]
- 2024 **Z. Li**, M. T. Jensen, A. Nolte, and D. Spikol. “Field report for Platform mBox: Designing an Open MMLA Platform.” *Proceedings of the 14th Learning Analytics and Knowledge Conference (LAK ’24)*, <https://doi.org/10.1145/3636555.3636872> [Oral presentation]

GRANTS AND AWARDS

- 2026 Meta Project Aria Research Partnership / Hardware Grant
- 2026 Best Full Paper Nomination @ ICALT ’26
- 2025 Best Short Paper Award @ LAK ’25
- 2024–2027 PhD Fellowship, University of Copenhagen

PROFESSIONAL EMPLOYMENT

- May 2019–Aug 2019 Yonyou, IT Consultant, Hong Kong, China
- Jun 2017–Aug 2017 Sina.com, Data Analyst Intern, Beijing, China

SERVICE

Peer reviewer: *Knowledge-Based Systems; European Conference on Technology-Enhanced Learning (EC-TEL)*

Updated September 2026